



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Determine the Whole Given the Part and Percent

Lesson 4-5 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can determine the whole when I know a part and the percent that part represents.

Language Objective: I can explain how I found the whole using the words part, whole, percent, and equation.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Determine the Whole Given the Part and Percent

Part

Whole

Double number line

Equation

Percent



Tap any word to see what it means and a picture.

1

Working backwards from a part and its percent finds the

2

How much of the base you get — your answer — is the

3

The full amount a percent is measured against is always

 %

4

Two parallel lines pairing amounts with their percents form a

5 The percent number sentence is part = × .

6 The word that means “per hundred” is .

Write About the Math

What did the car cost when they bought it, and how does your method show it?

¿Cuánto costó el carro cuando lo compraron, y cómo lo muestra tu método?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Determine the Whole Given the Part and Percent**

Hallar el total dados la parte y el porcentaje

☐ **Part**
Parte

☐ **Whole**
Total

☐ **Double number line**
Recta numérica doble

☐ **Equation**
Ecuación

Model: The original price is the whole (100%), and \$42 is only part of it (60%), so the whole must be more than \$42. — Students reason from $60\% < 100\%$ to whole $>$ part, without computing anything.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The \$26,600 is the ____ and the purchase price is the _____. Written as an equation, $0.7v =$ _____. Dividing both sides by _____ gives $v =$ _____. The whole is _____ than the part, which makes sense because 70% is less than 100%.
 - My answer is _____ because _____.
Mi respuesta es _____ porque _____.
-

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is _____.
Mi afirmación es _____.

- **My evidence is** ____, **so** ____.
Mi evidencia es ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Determine the Whole Given the Part and Percent
2. **Notes 2:** Part
3. **Notes 3:** Whole
4. **Notes 4:** Double number line
5. **Notes 5:** Equation
6. **Notes 6:** Percent
7. **Watch for this mistake:** *A common mistake when finding the whole is multiplying by the percent instead of dividing. For "\$42 is 60% of the original price," computing $0.6 \times 42 = \$25.20$ makes the original price SMALLER than the sale price — impossible for a discount. Since $\text{percent} \times \text{whole} = \text{part}$, the whole is found by dividing: $42 \div 0.6 = \$70$. Whenever the percent is less than 100%, the whole must come out larger than the part, so a smaller answer is the tell.*
8. **Try It 1:** — *"Of what number?" signals a missing whole. "What is ___% of ___?" hands you the whole and asks for the part.*
9. **Try It 2:** — *Every one is the same move: divide the part by the percent written as a decimal.*